Regime-mixture exceedance probabilities from a partly observed path, with collapse diagnostics

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Abstract

`regimix` gives the probability that a process ends above a threshold, from a path whose spread switches between levels, by fitting a hidden Markov model (unobserved switching states) without labels and mixing per-state Gaussian tail probabilities. What it offers is engineering, not a better forecast: two fixes to hmmlearn 0.3.3 (`min_covar` is not applied in the M-step, so a variance silently collapses to 0.002 times the floor on a synthetic repro; spherical `covars_` has the wrong shape), a batched Baum–Welch that fits 100,000 short sequences 131× faster than hmmlearn with the same estimates, and diagnostics that warn when the regimes are not real. As a forecaster it ties a plain realised-volatility probit on all 3 public datasets (EUR/USD: +0.0024 nats, SE 0.0015) and loses to a HAR-style probit on EUR/USD (+0.0043, SE 0.0012). Against a one-state probit it wins only on EUR/USD (-0.0079 nats, SE 0.0014, on 13,078 held-out episodes; negative favours regimix); on equities that is a tie and on M4, where the diagnostics warn, a loss. Its outputs never improve gradient boosting, and the floor never binds on the public data, so pristine hmmlearn EM gives identical results there.

1 Problem

An episode i is a scalar process X observed up to a decision time t_i, a threshold k_i and a remaining horizon $\tau_i > 0$. The label is

$$y_i = \mathbf{1}\{X(t_i + \tau_i) > k_i\}, \tag{1}$$

with $\mathbf{1}\{\cdot\}$ the indicator of the event, and the goal is a calibrated $P(y_i = 1)$ from the path before t_i. The plain answer is a probit, a Gaussian tail probability whose width comes from the recent path; we expected it to fail when activity jumps between a quiet and a busy level faster than a realised-volatility window (summed squared increments) can follow.

A filtered regime, read off a few lookback windows, should react within one window. The filter is Hamilton's [Hamilton, 1989] and the regime-weighted normal tail is standard in regime-switching return models [Hardy, 2001]; we claim neither. The delta is two fixes to the upstream library [hmmlearn developers, 2024], a batched Baum–Welch over many short episodes, one label-free HMM on all their lookback windows, and diagnostics that say when the regimes are not real; the out-of-fold calibration is a design choice with no measured gain (Section 4). The expectation failed: a realised-volatility probit ties regimix on every dataset.

Preprint.

2 The change

Pick lookback edges $0 = e_0 < e_1 < \cdots < e_S$. Segment s is the increment over one window, scaled to unit time:

$$z_{i,s} = \mathrm{clip}\left(\frac{X(t_i - e_{s-1}) - X(t_i - e_s)}{\sqrt{e_s - e_{s-1}}},\, -c,\, c\right), \tag{2}$$

$$w_i = \alpha_{i,S}\, A, \qquad \alpha_{i,S}(j) = P(r_S = j \mid z_{i,1}, \ldots, z_{i,S}), \tag{3}$$

$$\tilde{p}_i = \sum_{j=1}^{K} w_i(j)\, \Phi\left(\frac{d_i + \mu_j \tau_i}{\max(\sigma_j, \sigma_{\min})\sqrt{\tau_i}}\right), \qquad d_i = X(t_i) - k_i, \tag{4}$$

$$p_i = \mathrm{sigmoid}\big(a + b\ \mathrm{logit}\, \tilde{p}_i\big). \tag{5}$$

Symbol	Meaning	Symbol	Meaning
$z_{i,s}$	segment s, oldest first	c	optional clip
r_s, K	regime at segment s; regime count	A	transition matrix
$\alpha_{i,S}$	filtered regime probabilities	w_i	horizon regime weights
μ_j, σ_j^2	drift, variance per unit time ($\mu_j = 0$)	$\sigma_{\min}$	spread floor (0.05)
Φ	normal CDF	a, b	Platt parameters [Platt, 1999, Lin et al., 2007]

We fit the hidden Markov model (HMM) with K Gaussian states by Baum–Welch [Baum et al., 1970] on the segment sequences of all episodes, without labels. Equation (3) is its forward pass, batched over episodes, moved one step ahead by A (next-step weighting), and Equation (4) averages one random-walk tail probability per regime. Labels enter only in Equation (5), through two parameters that we fit on outputs from 5 contiguous time-block folds before the HMM is refitted on all training rows. The estimator also returns w_i and $\mathrm{logit}\, \tilde{p}_i$, so that a downstream learner can use the regimes as inputs. We claim three properties, and Appendix A proves each of them.

Lemma 1 (exact mixture)**.** *If, given the horizon regime $r = j$, $X(t_i + \tau_i) - X(t_i) \sim \mathcal{N}(\mu_j \tau_i, \sigma_j^2 \tau_i)$ independently of $z_{i,1..S}$, $P(r = j \mid z_i) = w_i(j)$ and $\sigma_j \geq \sigma_{\min}$, then $\tilde{p}_i = P(y_i = 1 \mid z_i)$.*

Lemma 2 (no test label enters)**.** *If no training label reads a value of X inside the held-out period, then p_i for a held-out episode depends on labels only through training labels.*

The second part of the change is a variance floor: hmmlearn documents `min_covar` as a floor on the covariance but applies it only at initialisation, so on data with many exact zeros one state collapses onto a spike at zero. Its likelihood then grows without bound and its posterior takes only two values, a one-bit gate. Our `FlooredGaussianHMM` applies the floor after every M-step (the parameter-update half of EM):

$$v_j \leftarrow \max\big(v_j^{\mathrm{EM}},\, m\big)\,. \tag{6}$$

Here v_j^{EM} is the unconstrained variance update and m is `min_covar` (0.05 by default).

Lemma 3 (monotone EM)**.** *For diagonal or spherical covariances, update (6) is the exact maximiser of the M-step objective over $\{v \geq m\}$, so the log-likelihood never decreases.*

We turn the conditions under which regimes can help into warnings, computed from the fitted variances v_j and the self-transition probabilities A_{jj}:

$$\rho = \frac{\min_j v_j}{m}, \qquad D_j = \frac{1}{1 - A_{jj}}, \qquad R = \frac{\max_j \sigma_j}{\min_j \sigma_j}. \tag{7}$$

Here ρ is the collapse ratio (near 1: a state sits on the floor), D_j the mean dwell in segments and R the spread ratio. A dwell below 2 segments or a spread ratio below 2 triggers a warning; the report also gives the share of exactly flat segments, the count of distinct posterior values (a gate has two) and per-segment permutation importance. Appendix C gives the inputs, the four calls that apply regimix to any path, and the commands that reproduce every number here.

3 Setup

Data. We use 3 public datasets, each chosen because we expected its activity level to switch (on M4 the diagnostics later say it does not). *EUR/USD 1-minute* bars from HistData, 2016–2023: at a random

minute each hour, $P(\text{price in 30 min} > \text{price} + u)$ with $u \in \{-3, 0, +3\}$ pips (1 pip $= 0.0001$ in price). *US large-cap daily* closes for 59 tickers, 2005–2024, one decision day in 4 per ticker: $P(\text{close in 10 days} > \text{close} \times (1 + u))$, $u \in \{-2, 0, +2\}\,\%$. *M4 hourly*, the hourly series of the M4 forecasting competition [Makridakis et al., 2020], one decision in 9 hours: $P(\log y_{t+h} > \log y_t + u)$ for $h \in \{12, 24\}$ hours and $u = c\,\hat{\sigma}_1\sqrt{h}$ ($\hat{\sigma}_1$ the one-step training spread), $c \in \{-0.5, 0, 0.5\}$. Appendix B lists the lookback edges. The held-out set is the last 30 % of time (per series for M4) and is never shuffled; we drop every training row whose label reads into it, an embargo (a gap against look-ahead) of one horizon. That leaves 34,202 / 13,078, 51,360 / 22,125 and 51,952 / 23,288 training / held-out episodes, each series scaled by the spread of its one-step training increments.

Arms. All arms see the same training rows, segments, distances and horizons, and we give each a Platt map fitted on training rows. No arm gets a hyperparameter search: each runs fixed settings written down before the run. *regimix* runs its defaults ($K = 2$, `n_iter` 60, `tol` 0.01, floor and next-step weighting $w_i = \alpha_{i,S}A$ on, drift off), and the *one-regime pooled probit*, our trivial baseline, is the same model with $K = 1$: one constant spread. The *realised-vol probit* uses the realised variance per unit time over the lookback; the *HAR-style probit* (heterogeneous autoregressive volatility) regresses log realised variance over the horizon on its value at each lookback window [Corsi, 2009, Andersen et al., 2003]. The *gate ablation* replaces w_i by a one-hot weight on the high-spread state when $|z_{i,S}|$ exceeds the training quantile that matches that state's stationary share (its long-run fraction of segments under A). *Logistic regression* and *histogram gradient boosting* [Pedregosa et al., 2011] get $(Z, |Z|, d, \tau, d/\sqrt{\tau})$. *Pristine hmmlearn EM* is the upstream 0.3.3 M-step [hmmlearn developers, 2024] without the floor (our batched E-step gives the same parameters as upstream to a relative 2.4×10^{-13}, Appendix C), and the stacking test [Wolpert, 1992] appends out-of-fold w_i and $\operatorname{logit} \tilde{p}_i$ to the boosting inputs.

Metric and error. Log-loss in nats, averaged over 3 seeds; the paired difference per episode is

$$\Delta_i = \ell(p_i^{\text{regimix}}, y_i) - \ell(p_i^{\text{arm}}, y_i), \qquad \ell(p, y) = -y \log p - (1 - y) \log(1 - p). \tag{8}$$

Negative $\bar{\Delta}$ means regimix is better. Episodes in one time block share a path, so a per-episode SE is too small (FX against $K = 1$: 0.0011 against 0.0014). We therefore sum Δ_i within blocks and run a moving-block bootstrap [Künsch, 1989] (resampling runs of consecutive blocks) with 2,000 draws over 749 FX days and 1,500 equity dates, with the block length set by the first lag at which the autocorrelation of block means falls inside $2/\sqrt{n}$. M4 series are independent, so there we resample 414 whole series (a Newey–West SE [Newey and West, 1987] is kept as a check). We call a difference a tie when its 95 % interval covers zero. Calibration is the expected calibration error over 15 bins (ECE, the mean gap between predicted and observed frequency) [Guo et al., 2017]. The 15 arms of the main run were written down before any held-out number existed. After the main run lost to the volatility probits we tried 6 post-hoc variants on the same held-out period, about 18 comparisons, and kept a variant only if it won on all 3 datasets.

4 Results

No dataset gives regimix a win over the best volatility probit (Table 1). On EUR/USD the regimes pass both diagnostics: they persist 19.6 and 8.7 segments on average and their spreads differ 3.37-fold. There regimix beats the pooled probit by 0.0079 nats and the gate ablation by 0.0075. The realised-vol probit is a tie (+0.0024, interval -0.0005 to +0.0053) and the HAR probit wins by 0.0043 (+0.0019 to +0.0067). Our reading, not tested, is that two discrete states are a coarse copy of the continuous volatility level that the HAR arm estimates directly.

On equities (Figure 1) the regimes pass both warnings (dwell 14.9 / 3.9, spread ratio 2.72), yet every probit, the pooled one included, lands within 0.0008 nats of regimix: passing the diagnostics does not guarantee a gain. On M4 the diagnostics warn on both counts (dwell 1.0 / 1.3 segments, spread ratio 1.09), and M4 is exactly where regimix loses to $K = 1$, by 0.0029 nats. Against the HAR probit regimix gains -0.0007 there (interval -0.0013 to -0.00005), which by our rule is a narrow win; the realised-vol probit, the better probit on M4, is a tie (+0.0010, interval -0.0002 to +0.0020). On M4 logistic regression and boosting reach 0.3476 and 0.2126, far below every probit (untested reading: seasonality and mean reversion a driftless walk cannot express), so M4 ranks regimix only among probits.

Table 1: Regimix beats the pooled probit only on EUR/USD (a tie on equities, a loss on M4) and never beats the best volatility probit. Held-out log-loss, nats; in brackets regimix minus arm ± bootstrap SE (negative: regimix better). Last row: boosting with regime outputs minus boosting (positive: appending hurt).

arm	EUR/USD 1-min (n = 13,078)	US large-cap daily (n = 22,125)	M4 hourly (n = 23,288)
regimix ($K = 2$)	**0.6201**	**0.6505**	**0.5878**
pooled probit ($K = 1$)	0.6280 (-0.0079 ± 0.0014)	0.6508 (-0.0003 ± 0.0005)	0.5849 (+0.0029 ± 0.0002)
realised-vol probit	0.6177 (+0.0024 ± 0.0015)	0.6505 (0.0000 ± 0.0008)	0.5869 (+0.0010 ± 0.0006)
HAR-style probit	0.6158 (+0.0043 ± 0.0012)	0.6496 (+0.0008 ± 0.0005)	0.5885 (-0.0007 ± 0.0003)
gate ablation	0.6276 (-0.0075 ± 0.0012)	0.6541 (-0.0036 ± 0.0005)	0.5910 (-0.0032 ± 0.0003)
logistic regression	0.6288 (-0.0087 ± 0.0016)	0.6523 (-0.0018 ± 0.0008)	0.3476 (+0.2402 ± 0.0049)
gradient boosting	0.6195 (+0.0006 ± 0.0010)	0.6530 (-0.0025 ± 0.0008)	0.2126 (+0.3752 ± 0.0053)
pristine hmmlearn 0.3.3 EM	0.6201 (0.0000 ± 0.0000)	0.6505 (0.0000 ± 0.0000)	0.5878 (0.0000 ± 0.0000)
boosting + regime − boosting	+0.0004 ± 0.0004	+0.0011 ± 0.0008	+0.0015 ± 0.0004

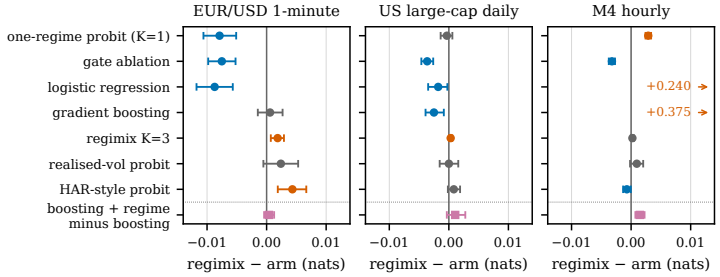


Figure 1: The best volatility probit matches or beats regimix on every dataset, and the regime outputs never help boosting. Paired held-out log-loss difference, regimix minus arm, 95 % block or (M4) series bootstrap intervals; blue a regimix win, red a loss, grey a tie, purple stacking.

Appended to boosting, the regime outputs tie on EUR/USD and equities and hurt on M4 (+0.0015, interval +0.0007 to +0.0023). Calibration follows the same order (Figure 2; ECE gaps untested): on EUR/USD the seed-averaged ECE is 0.0181 for regimix, 0.0237 for the pooled probit and 0.0122 for HAR, while on M4 regimix has the highest ECE of the three, 0.0623.

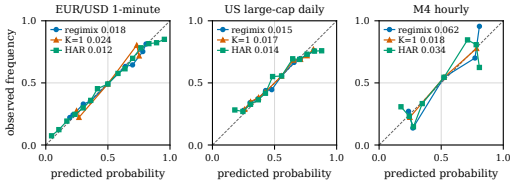


Figure 2: Regimix has a lower ECE than the pooled probit where regimes exist and a higher one on M4, where they do not. Reliability of the seed-averaged held-out probabilities, 15 bins; legend: ECE.

Among the ablations (Table 2) only $K = 3$ helps on two datasets (a tie on M4), still short of the volatility probits, and we keep $K = 2$ because $K = 3$ was picked on the held-out period; isotonic and per-regime Platt help only on M4. No post-hoc variant won on all 3 datasets. Out-of-fold against in-sample Platt, ablated after the main run, is of mixed sign (0.0000, -0.0004, +0.0008), so we credit it with no gain.

5 Limits

The idea lost a comparison and stays open: a realised-volatility probit reads the volatility level as well as the states do. To beat it, the states must carry what the level misses, such as a switch inside the horizon [Hardy, 2001], which regimix does not model. The evidence is thin: three datasets, one held-out period each, 3 seeds, post-hoc variants on that period, a floor and a speed-up measured only on synthetic data (one machine), survivorship bias in the equities, a drift unit mismatch (Appendix B), and no ablation of fitting one HMM on all episodes.

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A Proofs

Lemma 1. **Given:** 1) law of total probability over a partition $\{r = j\}_{j=1}^{K}$; 2) given $r = j$, the increment $U = X(t_i + \tau_i) - X(t_i) \sim \mathcal{N}(\mu_j \tau_i, \sigma_j^2 \tau_i)$, independent of z_i; 3) $P(r = j \mid z_i) = w_i(j)$; 4) $\Phi(-x) = 1 - \Phi(x)$.

Choose: the event $E = \{y_i = 1\} = \{X(t_i) + U > k_i\} = \{U > -d_i\}$ and the partition of rule 1.

Need to prove: $P(E \mid z_i) = \sum_j w_i(j)\, \Phi\big((d_i + \mu_j \tau_i)/(\sigma_j \sqrt{\tau_i})\big)$.

$$
\begin{aligned}
P(E \mid z_i) &= \textstyle\sum_j P(E \mid r = j, z_i)\, P(r = j \mid z_i) && \text{rule 1} \\
&= \textstyle\sum_j P(E \mid r = j)\, w_i(j) && \text{independence in rule 2, rule 3} \\
P(E \mid r = j) &= P\Big(\tfrac{U - \mu_j \tau_i}{\sigma_j \sqrt{\tau_i}} > \tfrac{-d_i - \mu_j \tau_i}{\sigma_j \sqrt{\tau_i}}\Big) && \text{standardise } U\text{, rule 2} \\
&= 1 - \Phi\Big(\tfrac{-d_i - \mu_j \tau_i}{\sigma_j \sqrt{\tau_i}}\Big) && \text{definition of } \Phi \\
&= \Phi\Big(\tfrac{d_i + \mu_j \tau_i}{\sigma_j \sqrt{\tau_i}}\Big) && \text{rule 4}
\end{aligned}
$$

Precondition: Equation (4) uses $\max(\sigma_j, \sigma_{\min})$, which equals σ_j because $\sigma_j \geq \sigma_{\min}$. **Therefore,** substitution gives $\tilde{p}_i = P(y_i = 1 \mid z_i)$. □

Lemma 2. **Given:** 1) $\theta = \mathrm{fit}(Z_{\text{train}})$, Baum–Welch on segments only; 2) $(a, b) = \mathrm{Platt}\big(\{\mathrm{logit}\, \tilde{p}_j^{\text{oof}}, y_j\}_{j \in \text{train}}\big)$, where each out-of-fold output comes from an HMM fitted on segments only; 3) for a held-out episode, $p_i = g(z_i, d_i, \tau_i; \theta, a, b)$ by Equations (3)–(5).

Choose: a held-out episode i and write p_i through the fitted quantities of rules 1 and 2.

Need to prove: p_i depends on labels only through $\{y_j\}_{j \in \text{train}}$.

$$
\begin{aligned}
p_i &= g(z_i, d_i, \tau_i;\, \theta,\, a,\, b) && \text{rule 3} \\
&= g\big(z_i, d_i, \tau_i;\, \mathrm{fit}(Z_{\text{train}}),\, \mathrm{Platt}(\{\mathrm{logit}\, \tilde{p}_j^{\text{oof}}, y_j\}_{j \in \text{train}})\big) && \text{rules 1 and 2}
\end{aligned}
$$

The only label symbols on the right are y_j with $j \in$ train.

Precondition: a training label must not itself carry held-out information. A training label reads $X(t_j + \tau_j)$; rows with $t_j + \tau_j$ inside the held-out period are dropped (the embargo of Section 3), so every remaining y_j is a function of X before the held-out period.

Therefore, no held-out label enters p_i. The out-of-fold step is not needed for this; it only makes the calibrator see raw outputs from an HMM that did not fit those rows, as at test time. □

Lemma 3. **Given:** 1) generalised EM [Dempster et al., 1977]: if each update satisfies $Q(\theta^{\text{new}} \mid \theta^{\text{old}}) \geq Q(\theta^{\text{old}} \mid \theta^{\text{old}})$, the log-likelihood does not decrease; 2) for diagonal or spherical covariances Q separates into one term per state and coordinate, and the variance term has the form $q(v) = -\frac{1}{2}(\nu \log v + s/v)$ with weight $\nu > 0$ and scatter $s > 0$ (a conjugate variance prior only changes ν and s); 3) the mean update does not depend on v.

Choose: one variance term q and the feasible set $v \geq m$.

Need to prove: $\arg\max_{v \geq m} q(v) = \max(s/\nu,\, m)$.

$$
\begin{aligned}
q'(v) &= -\tfrac{1}{2}\left(\tfrac{\nu}{v} - \tfrac{s}{v^2}\right) && \text{derivative of rule 2} \\
&= \tfrac{\nu}{2v^2}\left(\tfrac{s}{\nu} - v\right) && \text{factor out } \tfrac{\nu}{2v^2} > 0
\end{aligned}
$$

So $q' > 0$ for $v < s/\nu$ and $q' < 0$ for $v > s/\nu$: q is unimodal with peak $v^* = s/\nu = v^{\text{EM}}$. Same method on each side of m. **Case** $v^* \geq m$**:**

$$
\begin{aligned}
\arg\max_{v \geq m} q(v) &= v^* && \text{peak is feasible, unimodal } q \\
&= \max(v^*, m) && v^* \geq m
\end{aligned}
$$

Case $v^* < m$**:**

$$
\begin{aligned}
\arg\max_{v \geq m} q(v) &= m && q' < 0 \text{ on } [m, \infty) \text{ since } m > v^* \\
&= \max(v^*, m) && v^* < m
\end{aligned}
$$

Substitution gives update (6) in both cases.

Precondition: rule 1 needs θ^{old} feasible, so that it is a candidate of the constrained maximisation. The initial variances carry $+m$, so they are feasible. Every later variance is the output of (6), so it is $\geq m$. By induction every iterate is feasible.

Therefore, by rule 3 the means keep their usual update, each variance gets its exact constrained maximiser, Q does not decrease, and by rule 1 neither does the log-likelihood. □

B Ablations and post-hoc variants

The benchmark arm equals the public `RegimeExceedanceClassifier` with a maximum logit difference of 0. The lookback edges are 0, 1, 3, 7, 15, 31, 63 minutes (EUR/USD), 0, 1, 3, 7, 15, 31 days (equities) and 0, 1, 2, 4, 8, 16, 32, 48 hours (M4). The tie on M4 for $K = 3$ has interval -0.0001 to +0.0005. Turning drift on is a large loss on equities and M4; behind `drift=True` the segment mean is drift per $\sqrt{\text{time}}$ while Equation (4) reads it per unit time. Neither floor binds on the public data: the smallest stock-EM variance is 4.8, 8.3 and 30.5 times the floor, so both ablations are bit-identical. The floor's evidence is synthetic. On 4,000 sequences of length 7 drawn as rounded $\mathcal{N}(0, 0.5^2)$ values (about 68 % exact zeros), stock `GaussianHMM` with $m = 10^{-3}$ ends with variances 0.163 and 2.3×10^{-6}, which is 0.002 times the floor.

Table 2: No single default carries the method, the floors never bind on public data, and calibration matters most where regimes are not real (M4). Regimix minus the ablated arm, log-loss in nats ± paired SE; a positive value means the ablated arm is better. The last two rows were added after the main run.

ablation	EUR/USD 1-min	US large-cap daily	M4 hourly
$K = 3$	+0.0019 ± 0.0006	+0.0003 ± 0.0001	+0.0002 ± 0.0001
pristine EM (no floor)	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
next step off	+0.0001 ± 0.0002	-0.0005 ± 0.0001	-0.0003 ± 0.0000
$\sigma_{\min}$ off	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
drift on	0.0000 ± 0.0001	-0.0093 ± 0.0028	-0.1028 ± 0.0024
isotonic calibration	0.0000 ± 0.0004	-0.0003 ± 0.0003	+0.0076 ± 0.0006
per-regime Platt	+0.0002 ± 0.0003	-0.0010 ± 0.0002	+0.0020 ± 0.0003
in-sample Platt	0.0000 ± 0.0000	-0.0004 ± 0.0001	+0.0008 ± 0.0000
no calibration	+0.0008 ± 0.0004	-0.0052 ± 0.0032	-0.0149 ± 0.0009

All 6 variants were tried after the main run, on the same held-out period. Values are variant minus the named arm, ± paired SE (negative: the variant is better). The level-relative variants divide segments and distance by the realised-vol or HAR level, so the regimes model only the change around it.

Table 3: Dividing by the volatility level turns regimix into a near-copy of the level arm, and more regimes do not help against it. No variant wins on all three datasets.

variant	against	EUR/USD 1-min	US large-cap daily	M4 hourly
$K = 4$	realised vol	-0.0003 ± 0.0012	+0.0001 ± 0.0008	+0.0031 ± 0.0007
$K = 4$	HAR	+0.0016 ± 0.0010	+0.0009 ± 0.0005	+0.0015 ± 0.0006
rv-relative, $K = 2$	realised vol	-0.0002 ± 0.0001	0.0000 ± 0.0000	+0.0022 ± 0.0003
rv-relative, $K = 3$	realised vol	-0.0005 ± 0.0002	0.0000 ± 0.0000	+0.0060 ± 0.0005
HAR-relative, $K = 2$	HAR	0.0000 ± 0.0001	0.0000 ± 0.0000	+0.0022 ± 0.0002
HAR-relative, $K = 3$	HAR	-0.0001 ± 0.0000	0.0000 ± 0.0000	+0.0011 ± 0.0002
$n = 300$	realised vol	+0.0035 ± 0.0015	+0.0051 ± 0.0011	-0.0041 ± 0.0007
$n = 1{,}000$	realised vol	+0.0021 ± 0.0015	-0.0016 ± 0.0007	-0.0026 ± 0.0006
$n = 3{,}000$	realised vol	+0.0032 ± 0.0017	-0.0004 ± 0.0008	-0.0028 ± 0.0006
$n = 300$	boosting	-0.0281 ± 0.0023	-0.1556 ± 0.0087	+0.1881 ± 0.0046

The only result we have outside these benchmarks follows. On a proprietary binary threshold-exceedance task, the regime-mixture output did not beat a single-regime realised-volatility probit on its own: block-CV log-loss +0.010 nats worse (paired block-bootstrap SE 0.012, 95 % CI [-0.013, +0.032], i.e. a tie within noise). Against stock hmmlearn on held-out log-loss, a zero-inflated emission gave +0.0001 (SE 0.0037), a per-position variance +0.0014 (SE 0.0012), and a Student-t emission -0.0040 (SE 0.0015, 95 % CI [-0.0068, -0.0009]; negative is better).

The small-n rows train on the most recent 300, 1,000 or 3,000 episodes. With few labels regimix beats boosting by a wide margin on EUR/USD and equities; against the realised-vol probit it loses or ties on EUR/USD and wins only at one equities size and on M4. M4 training rows are ordered by series, so its small-n rows test transfer to other series, not a learning curve.

C Applying regimix to new data, and reproduction

Inputs. One observed path (or many), the decision times, a threshold per decision and a horizon. Divide the path by the spread of its one-step increments on the training part, so that segments and distances are in unit scale. Pick lookback edges that roughly double, for example $0, 1, 3, 7, 15, 31$ steps, with the oldest covering a few horizons. Keep episodes in time order: the out-of-fold calibration needs a splitter of contiguous time blocks, and the held-out set must come after the training set with an embargo of one horizon.

Calls. `Z = segments_from_path(x, times, t, edges)` builds Equation (2); `clf = RegimeExceedanceClassifier(n_states=2).fit(Z, y, distance=d, horizon=h)` fits the HMM and the calibrator, with $d = X(t) - k$; `clf.predict_proba(Z, distance=d, horizon=h)` and `clf.regime_posterior(Z)` return p_i and w_i; `diagnostics.report(clf, Z, y=y, distance=d, horizon=h)` prints the warnings of Equation (7). If the report warns on dwell or spread, the regimes are not real and $K = 1$ or a realised-vol probit is the better choice, as on M4; a clean report is no promise of a gain, as on equities. Always run a realised-vol probit beside it: on our data it was the arm to beat. `examples/quickstart.py` runs all of this on synthetic data with a known answer.

Speed. `python -m bench.run_speed` times 10 EM iterations of a two-state HMM on synthetic sequences of 5 segments, single-threaded, with the batched fit and with hmmlearn's `fit` (same initialisation, same M-step). The batched fit is 41× faster on 1,000 sequences, 139× on 10,000 and 131× on 100,000 (3.2 s against 417 s), and its parameters differ from hmmlearn's by at most a relative 2.4×10^{-13} (`bench/results/speed.json`).

Reproduction. Package, benchmark and paper live in the `regimix` repository; the benchmark results in `bench/results/` are from commit `a6e9ca4`, and `speed.json` was added later without changing them. The whole delta to upstream is `patches/hmmlearn/0001-gaussianhmm-apply-min-covar-after-every-m-step.patch` (and a spherical-shape fix, `0002`) against hmmlearn 0.3.3, with repro tests in `patches/hmmlearn/tests`. `bash bench/run_all.sh` downloads the data and writes `bench/results/{fx,eq,m4}[_variants].json` (about 1 hour on 2 threads); `python -m paper.results.ablate_calib {fx,eq,m4}` writes the calibration ablation to `paper/results/calib_*.json`, `python results/make_numbers.py` regenerates every number in this paper and `python figures/make.py` both figures (Figure 2 reads the held-out predictions the benchmark stores outside git).